17.

PALEOMAGNETISM OF THE BYLOT BASINS: EVIDENCE FOR MACKENZIE CONTINENTAL TENSIONAL TECTONICS

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Fahrig, W.F., Christie, K.W., and Jones, D.L., Paleomagnetism of the Bylot basins: Evidence for Mackenzie continental tensional tectonics; in Proterozoic Basins of Canada, F.H.A. Campbell, editor; Geological Survey of Canada, Paper 81-10, p. 303-312, 1981.

Abstract

A number of apparently related Helikian sedimentary and volcanic basins in northern Canada are here referred to collectively as the Bylot basins. These include the Borden and the Fury and Hecla basins of northern Baffin Island, and the Thule Basin of Greenland. The basins developed in response to widespread rifting activity 1220 Ma ago which resulted from tensional tectonics of continental dimensions. Mackenzie magmatism, which took place throughout the western Canadian Shield, and basaltic magmatism in southern Greenland and Fennoscandia, appear to be related to the same tensional event. The precise time correlation of early Borden Basin development with Helikian basaltic magmatism throughout the Canadian Shield and throughout a reconstructed supercontinent is based chiefly on paleomagnetic evidence.

The lowest units of the Borden sequence, Nauyat volcanics and Adams Sound Formation, are normally magnetized and yield poles at 167.5°W, 4.5°N, δm = 13.0°, δp = 6.5° and 166.0°W, 1.5S, δm = 6.0°, δp = 3.0°. Their combined pole for 12 sites is at 166.5°W, 1.5°N, δm = 6.0°, δp = 3.0°. This is indistinguishable from Mackenzie igneous poles. The Strathcona Sound Formation, which occurs near the top of the Borden sequence, exhibits six reversals in the sequence sampled and has a pole at 155.5°W, 8.0°N, δm = 4.0°, δp = 2.0° which is distinguishable from the Mackenzie pole.

The Aston Formation, which occurs on Somerset and Prince of Wales islands, has a pole position at 178.5°W, 2.0°S, δm = 13.4°, δp = 6.7°, which suggests that it is the time equivalent of the lower Borden sequence.

The fill that makes up the Borden Basin was accumulated over a period of 18 Ma, commencing about 1220 million years ago.

Résumé

Dans cette étude, on désigne collectivement par le terme de bassins de Bylot un certain nombre de bassins sédimentaires et volcaniques datant de l'Hélikien, et sans doute apparentés les uns aux autres. Ils comprennent le bassin de Borden et celui de Fury et Hecla dans le nord de la terre de Baffin, et le bassin de Thulé au Groenland. Ces bassins se sont formées il y a 1220 Ma sous l'influence d'une vaste activité tectonique, les fractures résultant de tensions s'exerçant à une échelle continentale. L'épisode magmatique de Mackenzie, qui s'est exercé sur tout l'ouest du Bouclier canadien, et le magmatisme basaltique du sud du Groenland et de la Fennoscandie, ont probablement été engendrés par le même épisode de tensions. Pour établir une corrélation temporelle précise entre le développement initial du bassin de Borden et le magmatisme basaltique de l'Hélikien, sur l'ensemble du Bouclier canadien et le supercontinent original, on s'est principalement basé sur des indices paléomagnétiques.

Les unités les plus anciennes de la succession de Borden, c'est-à-dire les couches volcaniques de Nauyat et la formation d'Adams Sound, sont magnétisées de façon normale et comportent des pôles de coordonnées 167,5°W, 4,5°N, δm = 13,0°, δp = 6,5° et 166,0°W, 1,5S, δm = 6,0°, δp = 3,0°. En combinant les pôles de 12 sites, on obtient les coordonnées suivantes: 166,5°W, 1,5°N, δm = 6,0°, δp = 3,0°. Ces pôles se confondent parfaitement avec les pôles d'activité volcanique de Mackenzie. La formation de Strathcona Sound, que l'on rencontre près du sommet de la succession de Borden, présente 6 inversions de polarité dans la succession échantillonnée, et possède un pôle de coordonnées 155,5°W, 8,0°N, δm = 4,0°, δp = 2,0°, qui se confond parfaitement avec le pôle de Mackenzie.

La formation d'Aston, que l'on rencontre sur les îles de Somerset et du Prince de Galles, se caractérise par un pôle de coordonnées 178,5°W, 2,0°S, δm = 13,4°, δp = 6,7° ce qui suggère qu'elle est l'équivalent stratigraphique de la partie inférieure de la succession de Borden.

Le matériel sédimentaire qui constitue le bassin de Borden s'est accumulé pendant une période de 18 Ma qui a débuté il y a environ 1220 Ma.

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INTRODUCTION

The stratigraphy and development of the Borden sequence is dealt with in detail elsewhere in this volume so only the briefest description of Borden stratigraphy is given here to provide a setting for the paleomagnetic data and for the synthesis of these data. The Borden sequence was divided into two groups (Blackadar 1963, 1970; Lemon and Blackadar, 1963). The lower group, the Eqalulik, comprises two units, the Nauyat, largely volcanic, which is conformably overlain by the Adams Sound, consisting of buff to purple quartzite. The Adams Sound is thought to be largely fluvial and grades upward (Blackadar, 1970) into the lowest unit of the Uluksan Group. The Uluksan consists of several formations and is largely shallow marine. Jackson et al. (1980) have indicated unconformities at two levels within the Uluksan, but these are unlikely to represent major lapses in time.

Most of the material collected for paleomagnetic study was obtained from three formations: the Nauyat and Adams Sound (Eqalulik Group) and the Strathcona Sound (Uluksan Group). The Strathcona Sound is the second highest unit of the Uluksan Group. In the lower Strathcona sampled section it consists chiefly of medium to dark chocolate-red siltstone and fine sandstone. The poles for the three formations may be used to indicate the direction and amount of polar wandering that took place during almost the entire time involved in filling the Borden Basin.

NAUYAT FORMATION

The Nauyat Formation consists largely of thick basaltic flows, which in the major area of exposure south of Adams Sound, extend over an area of at least 300 km^2 (Fig. 17.1). Six flows were sampled in the section indicated. These consist of four flows above and two below a thin pink quartzite unit. The flows are all at least 10 m thick and some are several tens of metres thick. A seventh flow probably occurs below the sampled section. The weathered surfaces of the flows vary considerably in colour, being rust, brick red, dark red, purple or grey-green. Individual flows have a number of distinguishing features which were described by Galley (1978) from a section a few kilometres to the southwest. The flows vary considerably in composition even within the same flow, but are chiefly subalkaline tholeiites. Alkalis total 5 per cent, and silica about 52 per cent (Galley, 1978).

Sixty cores were drilled from seven Nauyat sites (31 to 37), with site 31 representing the highest exposed flow. The collection includes 55 cores from the 6 flows mentioned above and 5 cores from a pink quartzite layer (site 35).

Two specimens were cut from each core and the 110 specimens from the six volcanic sites were A.F. demagnetized in five or six steps. With the exception of site 33, specimen directions shifted systematically to a fairly flat westerly direction during the step cleaning. At fields of 500 or 600 oe the directions within the 5 sites are highly

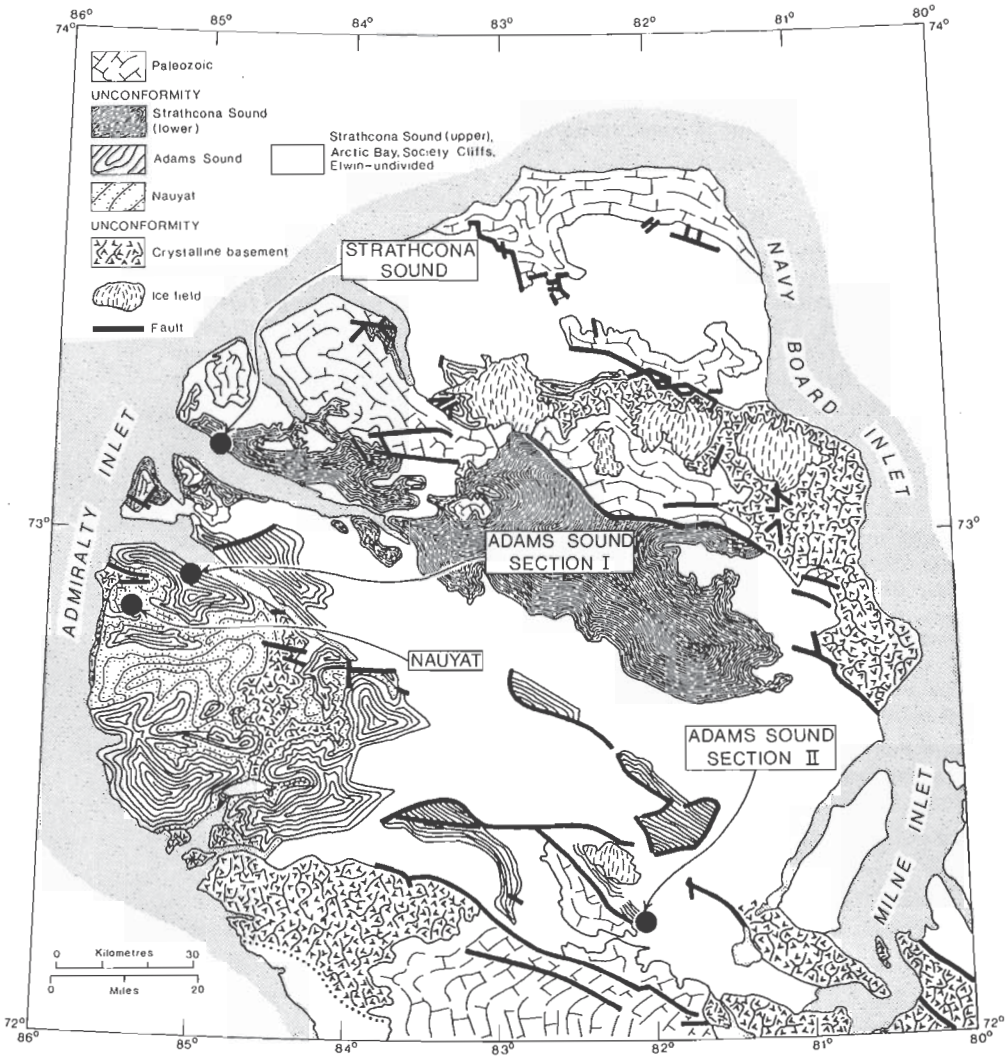


Figure 17.1

Geological map (simplified after Blackadar et al., 1968) showing location of sections sampled for paleomagnetic work. Only the basement gneisses, Paleozoic cover and the three formations sampled for paleomagnetic work, have been patterned.

dispersed. Specimen directions for site 33 showed the same westerly shift during step cleaning but were streaked towards the present earth's field direction. For this reason, data from this site were not included in group statistics for the formation (Table 17.1[1]). Figure 17.2 shows characteristic A.F. demagnetization curves for the six volcanic sites.

The pink quartzite (site 35) samples were thermally demagnetized in steps to 680°C. Sample directions were highly dispersed above 600°C and below this temperature retained a steeply down direction not far from that of the present earth's field. Figure 17.3 shows the demagnetization curves for one specimen of each core. These specimens were very weakly magnetic and it is unlikely that a primary magnetization was isolated during the thermal cleaning.

ADAMS SOUND FORMATION

The Adams Sound Formation was sampled at two localities (Fig. 17.1). At Section I, 36 cores were drilled at 7 sites through a 54 m very gently north-dipping sequence.

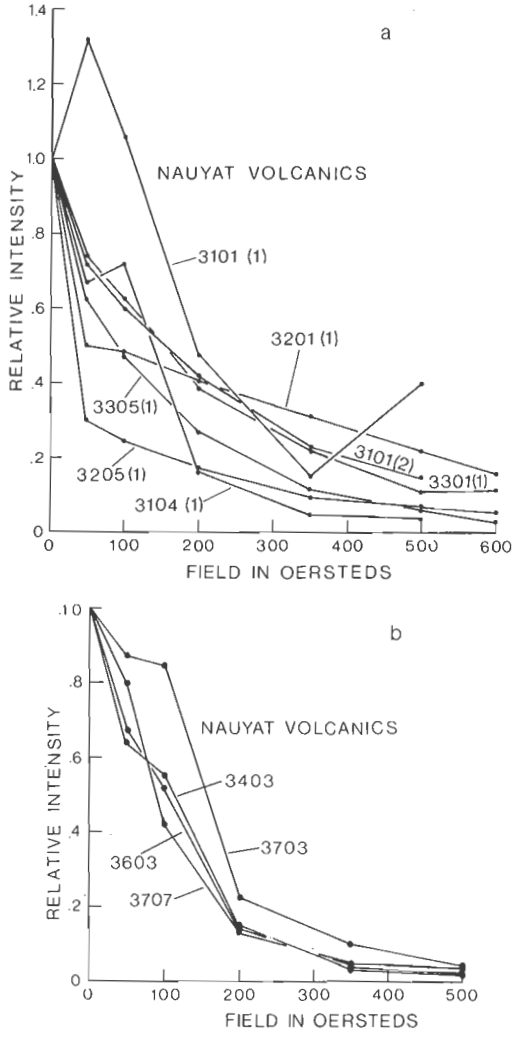


Figure 17.2. Alternating field demagnetization curves for Nauyat volcanics.

Site 38 is the highest site and site 44 the lowest. The sandstones here vary from buff to dark purple and drilling was concentrated in purple layers because it seemed likely that these would yield measurable cores.

One specimen from each core was thermally step cleaned and Figure 17.4 shows the response of these specimens to cleaning. The N.R.M. directions of the cores were well away from the earth's field direction (Table 17.2) and thermal demagnetization at 645°C caused a shift to a flatter westerly direction. The angles between the magnetization directions of specimens cut from the three cores of site 44 were all greater than 25° both before and after cleaning. Results for this site are not included in the formation mean even though their mean cleaned direction is very close to that of the remainder of the group.

A second set of cores (site 11) was collected at Adams Sound Section II (Fig. 17.1) about 130 km southeast of Section I. This collection consists of 17 cores drilled through a 15 m section. These cores were collected west of a 43 m thick diabase dyke. The core nearest the dyke was 76 m distant and the farthest core was about 300 m.

All of the specimens of this site were thermally cleaned in great detail. Their N.R.M. directions were steeply down and it was only as a result of cleaning at high temperatures that the majority of cores yielded the flat westerly directions characteristic of the cores from Section I (Fig. 17.5, Table 17.2). The difference in behaviour between these cores and those of Section I is ascribed to the proximity of Section II samples to the diabase dyke, which has a steeply down magnetization.

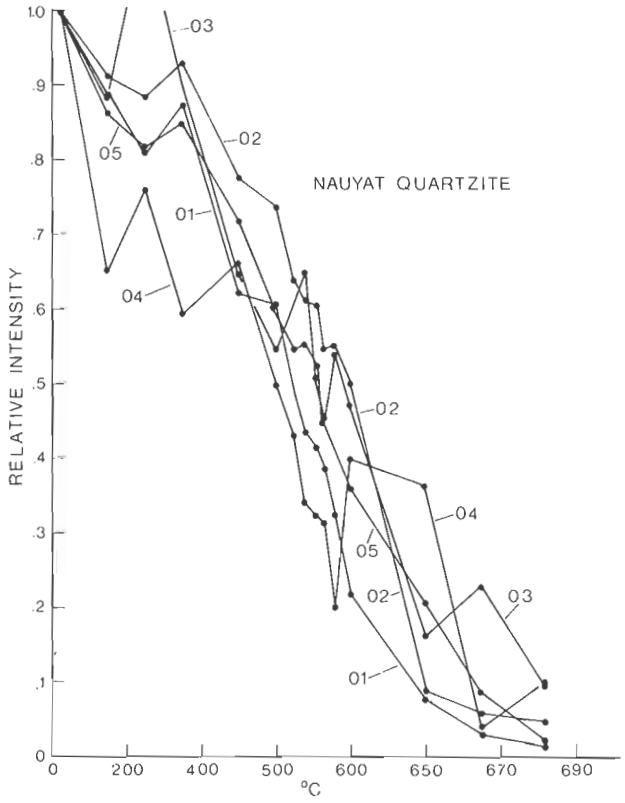


Figure 17.3. Thermal demagnetization curves for Nauyat quartzites.

[1] Tables for this paper appear after "Discussion"

The combined pole position for the Nauyat and Adams Sound formations (12 sites) is not statistically different from that for Mackenzie igneous rocks (Fahrig and Jones, 1969). The Adams Sound samples are all normally magnetized.

STRATHCONA SOUND FORMATION

Sixty-six samples were collected from a 245 m section in the Strathcona Sound Formation (Fig. 17.1). About 60 m of the section are repeated by faulting so the sampled stratigraphic interval is about 180 m. Site 5 was stratigraphically the highest and site 28 the lowest (Fig. 17.6). Sixty of the samples are from the lower Strathcona (Blackadar 1970) and the remaining six are from the upper Strathcona. The upper Strathcona at this locality is grey-green silty sandstone with some thin dark maroon layers. Site 5 consisted of this material and the three samples from this site proved unstable. Site 15, on the other hand, though similar in appearance, gave stable results. The lower Strathcona at this locality consists of rather friable red shales with layers of more massive dirty-red to chocolate-red silty sandstone. These more massive layers were sampled for paleomagnetic study.

Six clear polarity reversals were encountered in descending the section (Table 17.3, Fig. 17.6). The highest occurs several metres below the top of the lower Strathcona. This reversal is repeated by faulting (Fig. 17.6) and is referred to as the 'Strathcona Reversal'. Samples from the lower Strathcona were subjected to both alternating field and thermal demagnetization. The N.R.M. directions were well away from the present field direction and during cleaning their directions shifted only slightly. This shift took place in fields up to 400 oe and temperatures up to 400°C. Above these fields and temperatures no discernible changes in direction were detected (Fig. 17.7, 17.8). The site mean directions given in Table 17.3 are those obtained after cleaning at 600°C and 700 oersteds.

The magnetization direction of the Strathcona Sound is shown with that of the combined Adams Sound and Nauyat formations on Figure 17.10.

ASTON FORMATION

The Aston Formation was first mapped during "Operation Franklin" (Blackadar in Fortier et al., 1963). This is an unfossiliferous Proterozoic assemblage outcropping south of Aston Bay, Somerset Island. It consists predominantly of sandstone with minor shale and is overlain by the Hunting Formation, also Proterozoic, which is predominantly dolomite with minor shale and conglomerate. More detailed stratigraphic study of these formations has been carried out by Tuke et al. (1966), Dixon (1974) and Kerr (1977). Kerr indicated that the Hunting lies unconformably on the Aston.

Christie et al. (1971) correlated the basal sedimentary unit on the east coast of Prince of Wales Island with the Aston (Fig. 17.10, 17.11). In this area the unit consists predominantly of sandstone with conglomerate, dolomite and stromatolitic dolomite. It has been suggested that the depositional environment of the Aston ranged from supratidal to shallow subtidal. Paleomagnetic and K-Ar work on diabase from Savage Point suggested that the Aston is at least 1220 Ma old (Jones and Fahrig, 1978).

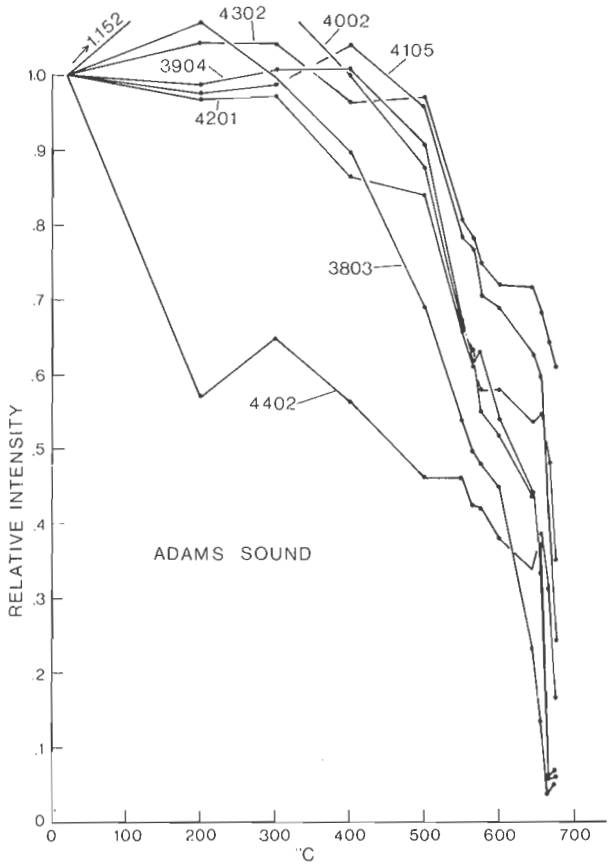


Figure 17.4. Thermal demagnetization curves for specimens of Adams Sound Formation from Section I (Fig. 17.1).

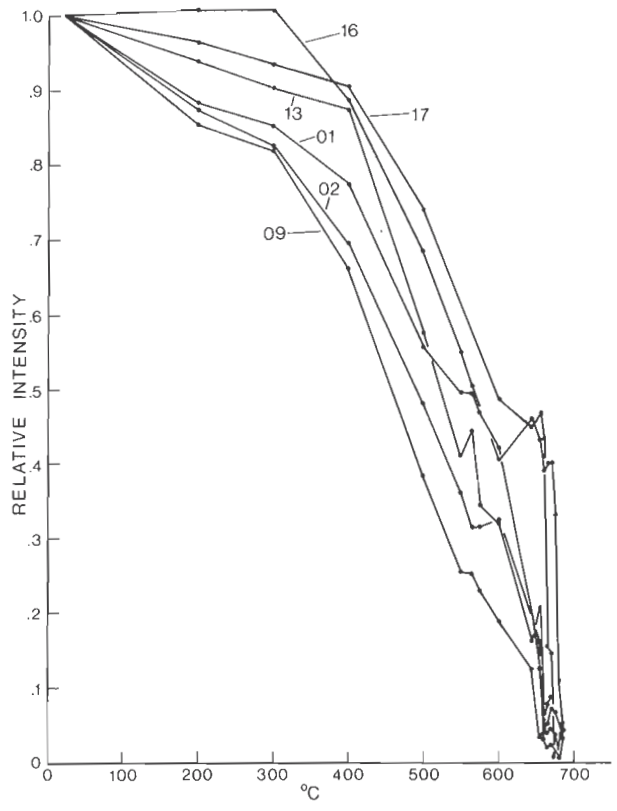


Figure 17.5. Thermal demagnetization curves for specimens of Adams Sound Formation from Section II (Fig. 17.1).

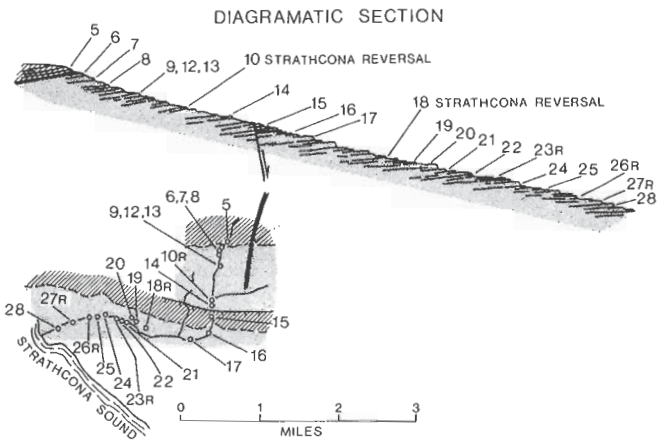


Figure 17.6. Below – plan view showing geology of lower Strathcona Sound Formation collecting site. Lower Strathcona stippled, upper Strathcona cross-hatched. "R" following site number indicates reversed polarity. Above – diagrammatic section indicating repetition of the Strathcona reversal by normal faulting.

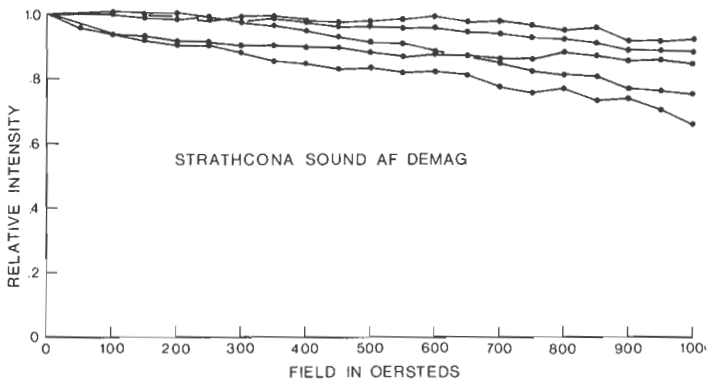


Figure 17.7. Alternating field demagnetization curves for lower Strathcona Sound Formation samples.

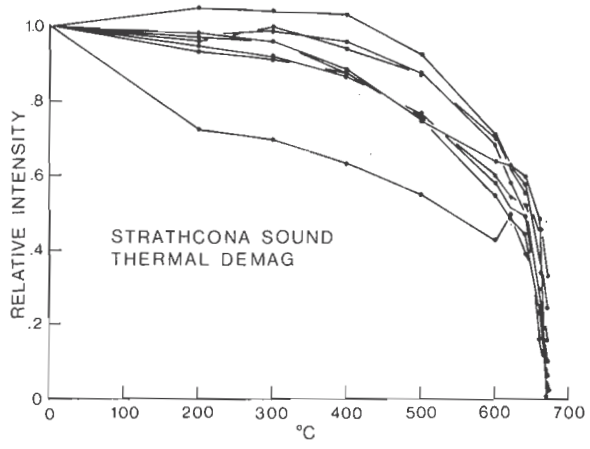


Figure 17.8. Thermal demagnetization curves for lower Strathcona Sound Formation samples.

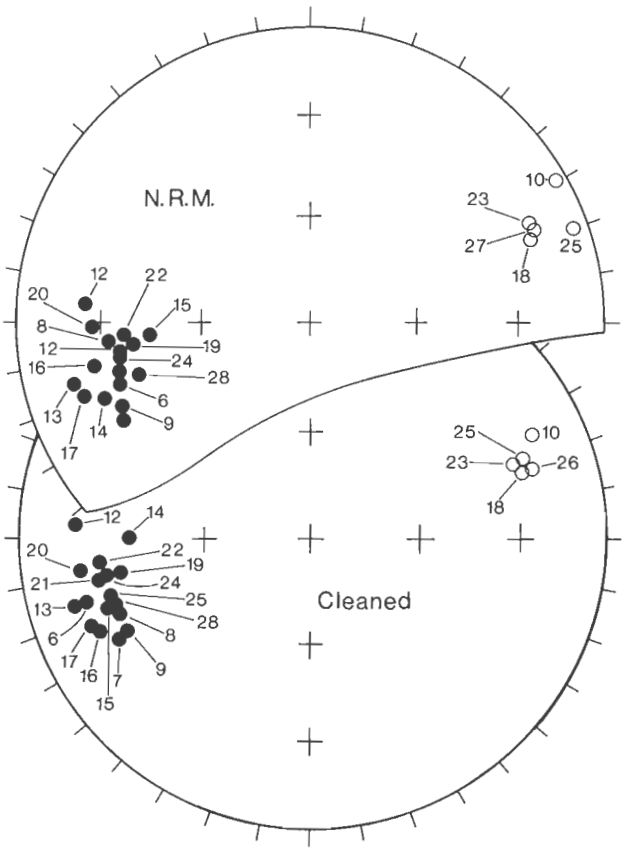


Figure 17.9. Equal area plot of site mean directions of Strathcona Sound before (upper) and after (lower) cleaning at 700 oersteds or 600° C.

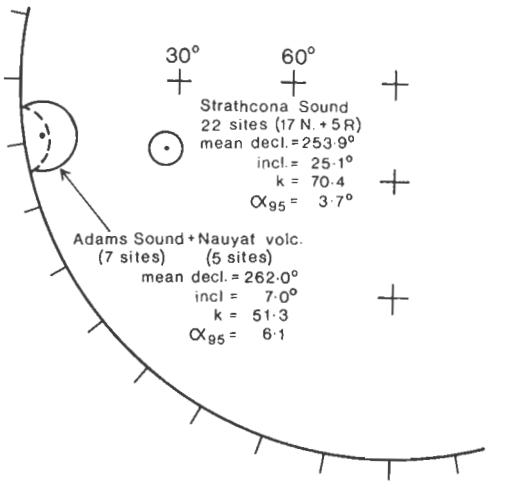


Figure 17.10. Stereographic (Wulff) plot of mean magnetization direction of the lower Borden sequence (Nauyat volcanics and Adams Sound Formation combined) and of the lower Strathcona Sound Formation.

Material for paleomagnetic work was collected from twelve sites near Aston Bay (Somerset Island) and Savage and Whitehead points (Prince of Wales Island). Results from only the five Savage Point sites are reported here (Table 17.4) because work was not completed on the remaining seven sites. Enough has been done, however, to indicate that at least some of the uncompleted sites will not give reliable paleomagnetic results.

Sites 3 and 5 consisted of well layered maroon silty sandstone, site 6 of buff to maroon feldspathic sandstone, and sites 12 and 13 of purple or maroon feldspathic sandstone.

DISCUSSION

The continent-wide nature of late Helikian Mackenzie igneous events was first established by Fahrig and Jones (1979) and was based chiefly on paleomagnetic evidence for the correlation of basic igneous rocks, particularly of diabase dykes, which suggested a remarkable period of North American crustal tension. The contemporaneity of the development of the Bylot rift basins with the Mackenzie events, as established in the present paper, extends evidence for this tension into the Arctic Islands (Fig. 17.13). In addition, Patchett et al. (1978) have pointed out that basic intrusive rocks in southern Greenland and Fennoscandia are very similar in age to the Mackenzie igneous rocks and have used the paleomagnetism of these rocks to reconstruct a North Atlantic supercontinent. Their reconstructed continent, to which the Bylot basins have been added, is shown in Figure 17.14.

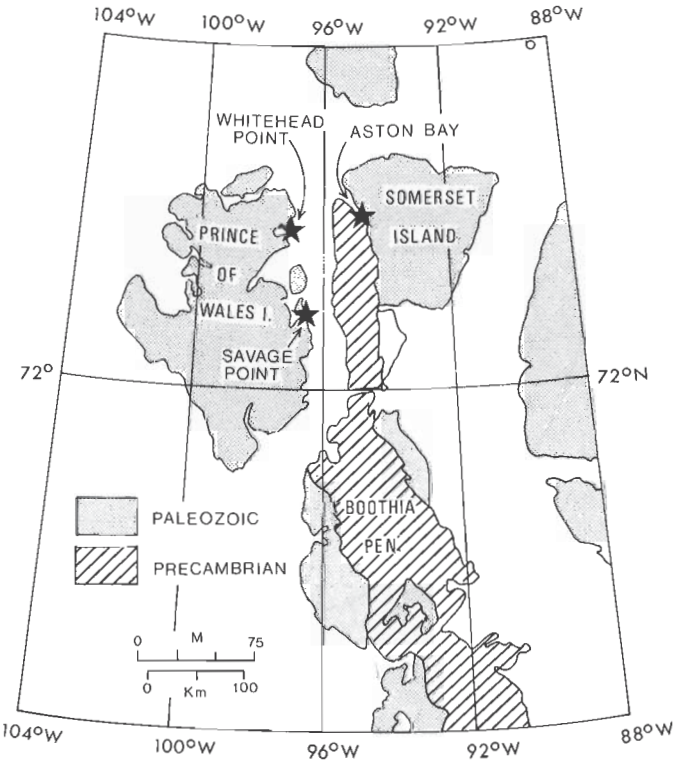


Figure 17.11. Index map showing the location of Aston Formation.

The correlation of Bylot basins with Mackenzie rocks has interesting implications regarding the Mackenzie Magnetostratigraphic Interval, which is a chronostratigraphic unit of the Helikian defined by Fahrig et al. (1971). At that time the upper and lower limits of the Interval could not be defined as virtually all the supporting paleomagnetic data were derived from intrusive rocks. The upper boundary of the Mackenzie Interval is now defined as the first reversal within the Borden sequence. At present the earliest known reversal is that in the lower Strathcona Sound Formation, so this is defined as the upper limit of the Mackenzie Magnetic Interval unless or until a reversal is discovered lower in the sequence.

As our knowledge of the importance of the Mackenzie tensional episode expands, support is provided for the suggestion by Douglas (1980) that the Mackenzie Magnetostratigraphic Interval be used as a fundamental marker in subdividing Helikian time.

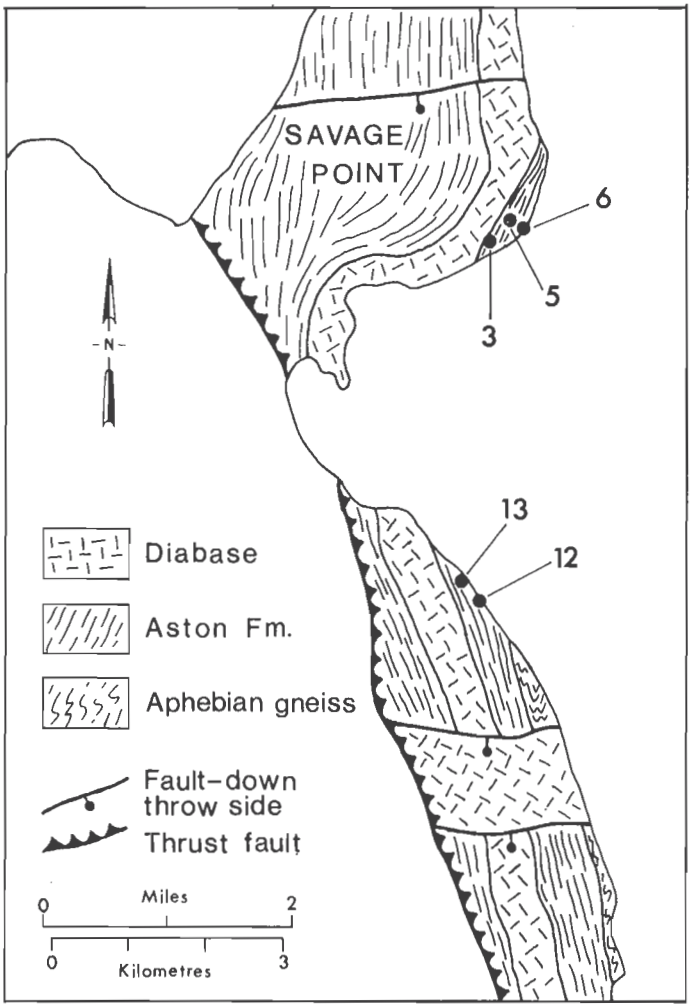


Figure 17.12. Local geology of the Savage Point area (after Christie et al., 1971) showing collecting sites of Aston Formation.

It should be possible from paleomagnetic data to estimate the time involved in filling the Borden Basin. The Mackenzie pole is the best dated of all North American Precambrian poles and the Logan dyke pole which forms a second anchor on the Logan Loop (Robertson and Fahrig, 1971) is fairly well dated at 1100 Ma. The intervening Logan Loop is the best defined part of the Precambrian North American polar wandering curve (Fig. 17.15). If it is assumed that the polar wandering rate between Nauyat volcanism and lower Strathcona sedimentation was the same as the average rate between 1220 and 1100 Ma, Nauyat to Strathcona time was 16.5 Ma. If 1.5 Ma are ascribed to upper Strathcona and Elwin sedimentation then the filling of the Borden Basin took 18 Ma.

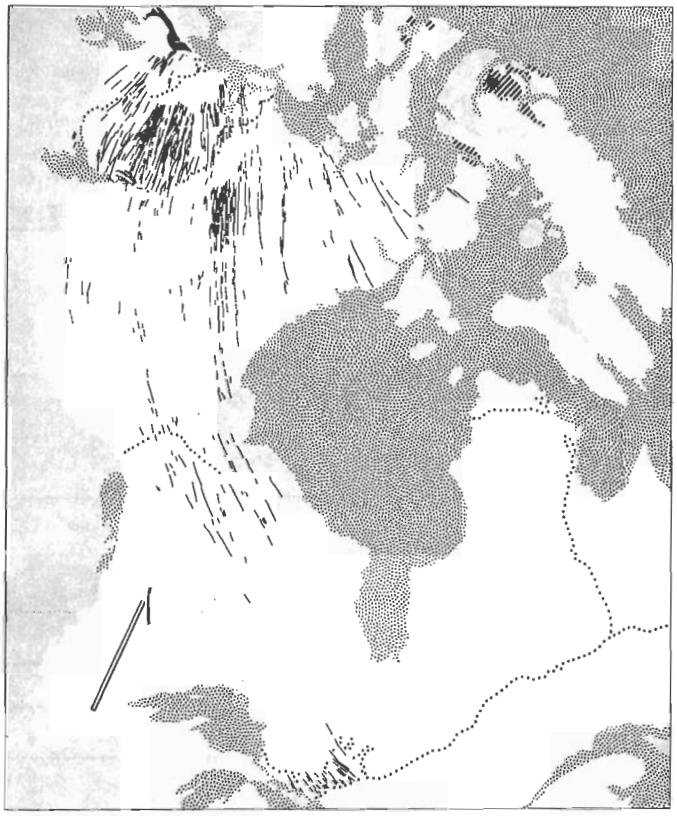


Figure 17.13. *Mackenzie igneous rocks; Coppermine volcanics, Muskox Complex, East Arm sills, Mackenzie dykes, Sudbury dykes and the Bylot basins (shown cross-hatched with the Nauyat volcanics in solid black; fine stippled areas are Paleozoic).*

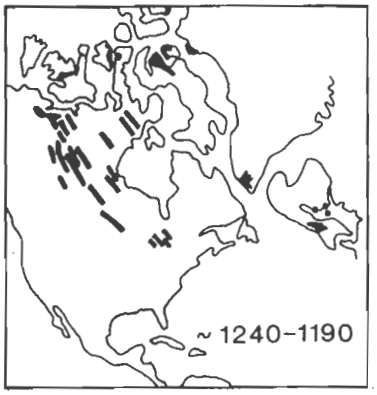


Figure 17.14. *Mackenzie supercontinent (Patchett et al., 1978) with the Bylot basins added to the Mackenzie igneous rocks.*

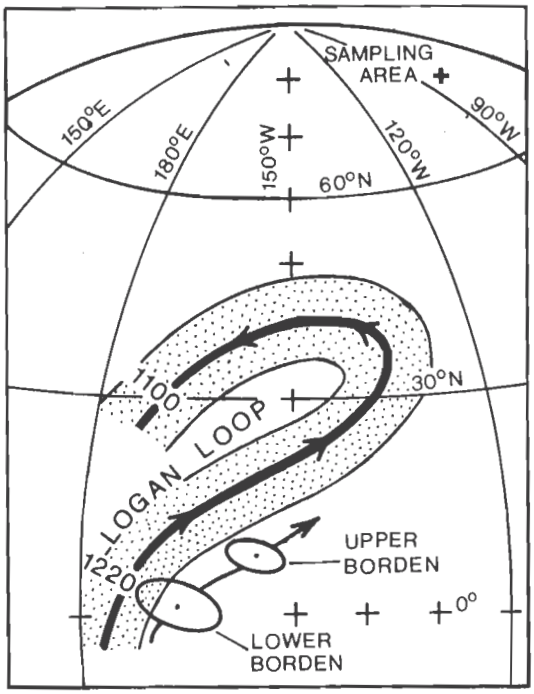


Figure 17.15. *Pole positions with ellipses of confidence for lower and upper Borden sequence.*

Table 17.1. Paleomagnetic data for Nauyat Volcanics

			NRM				CLEANED							
SITE	N	n	D	I	k	FIELD	D	I	k	α				
31	10	6	247.4	42.6	31.1	200 oe	261.0	22.4	45.3	10.1				
32	10	9	263.3	55.0	14.5	500 oe	273.4	0.4	50.5	7.3				
33*	9	9	264.6	66.1	99.4	200 oe	264.7	42.7	40.6	8.2				
34	9	9	240.3	50.6	26.1	200 oe	258.9	27.6	67.5	6.3				
36	10	10	254.7	27.6	32.4	200 oe	263.4	12.3	63.9	6.1				
37	7	7	256.8	17.7	93.8	350 oe	262.1	1.9	94.0	6.3				
											POLE FROM MEAN DIRECTION			
											LONG.	LAT.	δm	δp
SITES	6	5	252.7	39.0	23.1		263.9	13.0	37.2	12.7	167.7W	4.5N	13.0	6.6
SAMPLES	55	41	252.9	39.2	14.6		264.2	12.8	27.3	4.3	168.0W	4.5N	4.4	2.3

*Site 33 omitted from means and pole positions

N = number of samples collected at site
n = number of samples per site after rejecting unstable samples
D = declination (degrees)
I = inclination (degrees), negative numbers indicate north-seeking directions above the horizontal
k = Fisher's (1953) best estimate of precision
α = half angle (degrees) of the cone of confidence at P = 0.95
δm, δp = semi-axes of ellipse of confidence at P = 0.95

Table 17.2. Paleomagnetic data for Adams Sound Formation

			NRM				CLEANED							
SITE	N	n	D	I	k	TEMP.	D	I	k	α				
38	6	4	250.9	11.8	25.7	645°C	247.5	2.6	19.1	21.6				
39	5	5	255.0	9.1	51.6	645°C	252.9	-0.6	24.2	15.9				
40	5	5	264.2	17.6	47.5	645°C	262.2	6.8	91.3	8.1				
41	5	6	262.3	8.6	72.7	645°C	262.3	1.3	50.8	9.5	SECTION I			
42	6	4	261.7	15.7	222.9	645°C	262.8	5.2	181.2	6.8				
43	5	5	269.6	7.4	14.3	645°C	267.7	0.5	29.7	6.7				
44	3	0	-	-	-		-	-	-	-				
11	17	12	265.4	42.7	20.7	*645°C 675°C	264.2	5.3	25.6	8.7	SECTION II			
											POLE FROM MEAN DIRECTION			
											LONG.	LAT.	δm	δp
SITES	8	7	261.1	16.1	35.1		260.7	2.7	105.2	5.9	165.8W	1.4S	5.9	3.0
SAMPLES	53	41	262.0	20.4	15.4		261.7	3.1	32.7	4.0	166.7W	1.0S	4.9	2.0

* Optimum temperatures used for this site varied with sample:
1 at 645°, 3 at 655°, 3 at 660°, 4 at 665°, 1 at 675°.

Symbols as in Table 17.1.

Table 17.3. Paleomagnetic data for Strathcona Sound Formation

	NRM				CLEANED					
SITE	N	D	I	k	D	I	k	α	POLARITY	
6	1	251.9	32.8		254.5	20.2			N	
7	3	241.9	30.0	69.1	242.3	27.7	64.0	15.5	N	
8	4	264.2	32.5	67.5	248.8	29.8	29.1	17.3	N	
9	2	244.7	30.9		242.6	29.1			N	
10	2	59.5	-3.2		64.6	-18.5			R	
12	3	274.3	24.4	933.8	272.5	20.2	1546.6	3.1	N	
13	3	254.2	19.0	35.5	254.2	16.1	31.4	22.4	N	
14	1	248.5	26.7		269.8	37.5			N	
15	3	264.8	44.3	12.4	251.0	17.6	23.9	25.8	N	
16	3	257.3	26.3	10.2	246.5	22.1	17.3	30.6	N	
17	3	251.1	10.9	230.7	248.5	10.1	265.5	7.6	N	
18	4	69.2	-20.2	687.1	71.4	-26.4	464.0	4.3	R	
19	3	263.2	39.3	249.1	260.0	33.3	215.9	8.4	N	
20	2	268.9	28.1		262.3	20.8			N	
21	4	260.4	34.8	191.5	259.1	27.5	143.7	7.7	N	
22	4	264.9	38.0	171.7	263.5	27.6	326.7	5.1	N	
23	4	66.3	-18.9	75.1	69.7	-28.0	102.6	9.1	R	
24	3	258.8	34.8	212.6	259.6	28.3	134.8	10.7	N	
25	3	254.6	34.0	238.7	254.0	28.7	136.5	10.6	N	
26	3	70.2	-4.3	850.1	72.3	-22.9	240.7	8.0	R	
27	3	66.9	-18.3	3660.1	69.0	-25.1	915.8	4.1	R	
28	2	252.2	39.7		250.9	29.3			N	

									POLE FROM MEAN DIRECTION			
									LONG.	LAT.	δm	δp
N SITES	17	257.3	31.3	53.9	255.3	25.3	59.8	4.7	156.9W	8.5N	5.0	2.7
N SAMPLES	47	258.4	31.5	32.8	255.1	25.1	39.1	3.3	156.8W	8.4N	3.5	1.9
R SITES	5	66.4	-13.0	74.0	69.3	-24.2	311.8	4.3	151.6W	6.3N	4.6	2.5
R SAMPLES	16	67.0	-14.4	75.1	69.8	-24.9	172.2	2.8	151.8W	6.9N	3.0	1.6
ALL SITES	22	254.5	27.2	34.9	253.9	25.1	70.4	3.7	155.6W	8.0N	4.0	2.2
ALL SAMPLES	63	255.2	27.2	26.8	253.7	25.1	47.5	2.6	155.5W	8.0N	2.8	1.5

Polarity N = normal
R = reversed

Other symbols as in Table 17.1

Table 17.4. Paleomagnetic data for Aston Formation

	NRM						CLEANED				
SITE	N	n	D	I	k	TEMP.	D	I	k	α	
03	5	5	262.4	5.5	145.2	500°C	261.6	0.9	170.8	5.9	
05	10	9	261.4	6.4	132.1	500°C	259.2	1.2	122.9	4.7	
06	8	8	278.3	6.4	14.0	550°C	280.5	-4.5	18.6	13.2	
12	6	5	244.9	18.8	66.0	550°C	242.4	5.5	68.0	9.3	
13	8	8	268.9	15.0	20.0	550°C	264.1	1.3	11.6	18.5	

										POLE FROM MEAN DIRECTION			
										LONG.	LAT.	δm	δp
SITES	5	5	263.3	10.6	37.0		261.6	0.9	33.8	13.4	178.3°W	2.1°S	13.4	6.7
SAMPLES	37	35	264.7	10.2	20.6		263.0	0.5	18.6	5.9	179.7°W	1.9°S	5.9	2.9

Symbols as in Table 17.1

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